State Space Methods for Flight Vehicles

AE 695

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Distinct Eigenvalues

Let λ_i and $\boldsymbol{v}_i$ be the eigenvalue and corresponding eigenvector of matrix $\boldsymbol{A}$. Then, the set $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\}$ is linearly independent over $\mathbb{C}$.

- Proof by contradiction.
- Suppose set $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\}$ is linearly dependent.
- There exists $\alpha_i \ \forall i = 1, 2, \ldots, n$ (not all zero), s.t. $\sum_{i=1}^{n} \alpha_i \boldsymbol{v}_i = \boldsymbol{0}$
- Assuming $\alpha_1 \neq 0$, else reorder eigenvalues.
- Since $\boxed{(\boldsymbol{A} - \lambda_j \boldsymbol{I})\boldsymbol{v}_i = (\lambda_i - \lambda_j)\,\boldsymbol{v}_i \,\forall\, i \neq j \text{ and } (\boldsymbol{A} - \lambda_i \boldsymbol{I})\boldsymbol{v}_i = 0}$

$$(\boldsymbol{A} - \lambda_2 \boldsymbol{I})(\boldsymbol{A} - \lambda_3 \boldsymbol{I}) \cdots (\boldsymbol{A} - \lambda_n \boldsymbol{I}) \left(\sum_{i=1}^{n} \alpha_i \boldsymbol{v}_i \right) = \boldsymbol{0}$$

$$\implies \alpha_1 (\lambda_1 - \lambda_2)(\lambda_1 - \lambda_3) \cdots (\lambda_1 - \lambda_n)\, \boldsymbol{v}_1 = 0$$

- As λ_i are distinct, $\alpha_1 = 0$.
- A contradiction, and hence the set is l.i. and also qualifies for the basis.



Distinct Eigenvalues

- Let $\hat{\boldsymbol{A}}$ be the representation of $\boldsymbol{A}$ w.r.t. the basis $\{\boldsymbol{v}_i\}$.
- The ith column of $\hat{\boldsymbol{A}}$: Representation of $\boldsymbol{A}\boldsymbol{v}_i = \lambda_i \boldsymbol{v}_i$ w.r.t. $\{\boldsymbol{v}_i\}$.
- Representation of $\boldsymbol{A}$ w.r.t. the basis $\{\boldsymbol{v}_i\}$

$$\hat{\boldsymbol{A}} = \begin{bmatrix} \lambda_1 & 0 & 0 & \cdots & 0 \\ 0 & \lambda_2 & 0 & \cdots & 0 \\ 0 & 0 & \lambda_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_n \end{bmatrix}$$

- Using similarity transformation and $\boldsymbol{Q} = \begin{bmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \dots & \boldsymbol{v}_n \end{bmatrix}$

$$\boldsymbol{A}\boldsymbol{Q} = \boldsymbol{A}\begin{bmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \dots & \boldsymbol{v}_n \end{bmatrix} = \begin{bmatrix} \lambda_1\boldsymbol{v}_1 & \lambda_2\boldsymbol{v}_2 & \dots & \lambda_n\boldsymbol{v}_n \end{bmatrix} = \boldsymbol{Q}\hat{\boldsymbol{A}}$$

$$\boxed{\hat{\boldsymbol{A}} = \boldsymbol{Q}^{-1}\boldsymbol{A}\boldsymbol{Q}}$$

- For distinct eigenvalues of linear operator $\boldsymbol{A}$ that maps $(\mathbb{C}^n, \mathbb{C})$ to itself, operator has diagonal representation w.r.t. a set of eigenvectors as a basis.



Repeated Eigenvalues

- Consider the matrix given by

$$\boldsymbol{A} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

- Eigenvalues: $1, 1, 2$
- Eigenvectors:
  $\boldsymbol{v}_1 = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}^\top, \boldsymbol{v}_2 = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}^\top, \boldsymbol{v}_3 = \begin{bmatrix} -1 & 0 & 1 \end{bmatrix}^\top$
- L.i. eigenvectors, so qualify for a basis, and thus diagonalizable
- What about the matrix given below?

$$\boldsymbol{A} = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 2 \end{bmatrix}$$

- Eigenvalues: $1, 1, 2$
- Eigenvectors:
  $\boldsymbol{v}_1 = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}^\top, \boldsymbol{v}_2 = \begin{bmatrix} 5 & 3 & 1 \end{bmatrix}^\top$
- Only two l.i. eigenvectors, so they cannot qualify for the basis.



Repeated Eigenvalues

- Can we diagonalize a matrix having repeated eigenvalues?
- Not always in diagonal form, but representation in almost diagonal form

Jordan canonical form

A matrix is said to be in Jordan canonical form if its principal diagonal consists of Jordan blocks and the remaining elements are zero.

$$\begin{bmatrix} \lambda & 1 & 0 & \dots & 0 & 0 \\ 0 & \lambda & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & \lambda & 1 \\ 0 & 0 & 0 & \dots & 0 & \lambda \end{bmatrix}$$

- **Jordan form**: Eigenvalues on the diagonal and either 0 or 1 on its super-diagonal.
- What would be the different forms of Jordan representation with eigenvalue λ_1 with multiplicity 4 and λ_2 with multiplicity 1?
- Diagonal form is a special case of Jordan form.



Generalized Eigenvectors

A vector $\boldsymbol{v}$ is said to be generalized eigenvector of grade k of $\boldsymbol{A}$ iff

$$(\boldsymbol{A}-\lambda\boldsymbol{I})^{k}\,\boldsymbol{v}=\boldsymbol{0},\quad (\boldsymbol{A}-\lambda\boldsymbol{I})^{k-1}\,\boldsymbol{v}\neq\boldsymbol{0}$$

- Chain of generalized eigenvectors $\{\boldsymbol{v}_i\}$

$$\begin{aligned}
\boldsymbol{v}_k &\triangleq \boldsymbol{v}\\
\boldsymbol{v}_{k-1} &\triangleq (\boldsymbol{A}-\lambda\boldsymbol{I})\,\boldsymbol{v}=(\boldsymbol{A}-\lambda\boldsymbol{I})\,\boldsymbol{v}_k\\
\boldsymbol{v}_{k-2} &\triangleq (\boldsymbol{A}-\lambda\boldsymbol{I})^{2}\,\boldsymbol{v}=(\boldsymbol{A}-\lambda\boldsymbol{I})\,\boldsymbol{v}_{k-1}\\
&\cdots\cdots\\
\boldsymbol{v}_1 &\triangleq (\boldsymbol{A}-\lambda\boldsymbol{I})^{k-1}\,\boldsymbol{v}=(\boldsymbol{A}-\lambda\boldsymbol{I})\,\boldsymbol{v}_2
\end{aligned}$$

- $\mathcal{N}_i$: Null space of $(\boldsymbol{A}-\lambda\boldsymbol{I})^{i}$: All vectors $\boldsymbol{x}$ s.t. $(\boldsymbol{A}-\lambda\boldsymbol{I})^{i}\,\boldsymbol{x}=\boldsymbol{0}$
- Every vector in $\mathcal{N}_i$ is in $\mathcal{N}_{i+1}$, $\mathcal{N}_i\subset\mathcal{N}_{i+1}$
- Similarly, $\boldsymbol{v}_i=(\boldsymbol{A}-\lambda\boldsymbol{I})^{k-i}\,\boldsymbol{v}\in\mathcal{N}_i$, but $\boldsymbol{v}_i\notin\mathcal{N}_{i-1}$

$$(\boldsymbol{A}-\lambda\boldsymbol{I})^{i}\,\boldsymbol{v}_i=(\boldsymbol{A}-\lambda\boldsymbol{I})^{i}\,(\boldsymbol{A}-\lambda\boldsymbol{I})^{k-i}\,\boldsymbol{v}=(\boldsymbol{A}-\lambda\boldsymbol{I})^{k}\,\boldsymbol{v}=\boldsymbol{0}$$

$$(\boldsymbol{A}-\lambda\boldsymbol{I})^{i-1}\,\boldsymbol{v}_i=(\boldsymbol{A}-\lambda\boldsymbol{I})^{i-1}\,(\boldsymbol{A}-\lambda\boldsymbol{I})^{k-i}\,\boldsymbol{v}=(\boldsymbol{A}-\lambda\boldsymbol{I})^{k-1}\,\boldsymbol{v}\neq\boldsymbol{0}$$



- Consider an $n \times n$ matrix with eigenvalue λ with multiplicity m.
- How to generate the generalized eigenvectors?
- Process of searching a chain of generalized eigenvectors
- Compute ranks of $(\boldsymbol{A} - \lambda \boldsymbol{I})^i$ (that is, $\boldsymbol{B}^i$) for all $i = 0, 1, 2, \ldots$ until rank is $n - m$.
- Let us assume $n = 10$, $m = 8$, $k = 4$

$$\mathcal{N}_0 \subset \mathcal{N}_1 \subset \mathcal{N}_2 \subset \cdots \implies 0 = \nu_0 \leq \nu_1 \leq \nu_2 \leq \cdots \leq \nu_k = m$$

$\rho\left(\mathbf{A} - \lambda \mathbf{I}\right)^0 = 10, \quad \nu_0 = 0$
$\rho\left(\mathbf{A} - \lambda \mathbf{I}\right) = 7, \quad \nu_1 = 3$
$\rho\left(\mathbf{A} - \lambda \mathbf{I}\right)^2 = 4, \quad \nu_2 = 6$
$\rho\left(\mathbf{A} - \lambda \mathbf{I}\right)^3 = 3, \quad \nu_3 = 7$
$\rho\left(\mathbf{A} - \lambda \mathbf{I}\right)^4 = 2, \quad \nu_4 = 8$

$\nu_4 - \nu_3 = 1$	$\nu_3 - \nu_2 = 1$	$\nu_2 - \nu_1 = 3$	$\nu_1 - \nu_0 = 3$
		$\mathbf{w}_2 = \mathbf{w}$	$\mathbf{w}_1 = \mathbf{B}\mathbf{w}$
		$\mathbf{v}_2 = \mathbf{v}$	$\mathbf{v}_1 = \mathbf{B}\mathbf{v}$
$\mathbf{u}_4 = \mathbf{u}$	$\mathbf{u}_3 = \mathbf{B}\mathbf{u}$	$\mathbf{u}_2 = \mathbf{B}^2\mathbf{u}$	$\mathbf{u}_1 = \mathbf{B}^3\mathbf{u}$

$\mathcal{N}_1$
$\mathcal{N}_2$
$\mathcal{N}_3$
$\mathcal{N}_4$

No. of independent vectors in $\mathcal{N}_i$ but not in $\mathcal{N}_{i-1}$
Two chains with length 2
One chain with length 4



Generalized eigenvectors of $\boldsymbol{A}$ associated with the same λ are l.i.

- **To show that**: L.i. of $\boldsymbol{u}_2, \boldsymbol{v}, \boldsymbol{w} \implies$ L.i. of $\boldsymbol{u}_1, \boldsymbol{v}_1, \boldsymbol{w}_1$
- Suppose $\boldsymbol{u}_1, \boldsymbol{v}_1, \boldsymbol{w}_1$ are l.d. then $\exists\, c_i \neq 0 \,\forall\, i = 1, 2, 3$ s.t.

$$c_1\boldsymbol{u}_1 + c_2\boldsymbol{v}_1 + c_3\boldsymbol{w}_1 = \boldsymbol{0} = c_1\boldsymbol{B}\boldsymbol{u}_2 + c_2\boldsymbol{B}\boldsymbol{v} + c_3\boldsymbol{B}\boldsymbol{w} = \boldsymbol{B}(c_1\boldsymbol{u}_2 + c_2\boldsymbol{v} + c_3\boldsymbol{w}) = \boldsymbol{B}\boldsymbol{y}$$

- $\boxed{\boldsymbol{B}\boldsymbol{y} = \boldsymbol{0} \implies \boldsymbol{y} = \boldsymbol{0}}$ as $y \in \mathcal{N}_1$
- $\boldsymbol{y} = \boldsymbol{0} \implies c_i = 0 \,\forall\, i$. A contradiction
- To show that $\{\boldsymbol{u}_i \,\forall i = 1, 2, 3, 4, \boldsymbol{v}_j, \boldsymbol{w}_j \,\forall\, j = 1, 2\}$ are l.i.

$$c_1\boldsymbol{u}_1 + c_2\boldsymbol{u}_2 + c_3\boldsymbol{u}_3 + c_4\boldsymbol{u}_4 + c_5\boldsymbol{v}_1 + c_6\boldsymbol{v}_2 + c_7\boldsymbol{w}_1 + c_8\boldsymbol{w}_2 = \boldsymbol{0}$$

- Using $\boldsymbol{B}^3 = (\boldsymbol{A} - \lambda\boldsymbol{I})^3$, $c_4\boldsymbol{B}^3\boldsymbol{u}_4 = \boldsymbol{0}$. As $\boldsymbol{B}^3\boldsymbol{u}_4 \neq \boldsymbol{0}$, $c_4 = 0$.
- Using $\boldsymbol{B}^2$, we get $c_3 = 0$. With $c_3 = c_4 = 0$, using $\boldsymbol{B}$, we get

$$c_2\boldsymbol{u}_2 + c_6\boldsymbol{v}_2 + c_8\boldsymbol{w}_2 = \boldsymbol{0} \implies c_2 = c_6 = c_8 = 0$$

- Using the l.i. of $\boldsymbol{u}_1, \boldsymbol{v}_1, \boldsymbol{w}_1$, $c_1 = c_5 = c_7 = 0$.

Generalized eigenvectors of $\boldsymbol{A}$ associated with different λ are l.i. (HW)



Jordan Form Representation

- What would be the representation of $\boldsymbol{A}$ w.r.t. $\boldsymbol{Q}$?

$$\boldsymbol{Q} = \{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3, \boldsymbol{u}_4, \boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{w}_1, \boldsymbol{w}_2, \boldsymbol{x}_1, \boldsymbol{x}_2\}$$

- First 4 columns are representation of $\boldsymbol{A}\boldsymbol{u}_i$ w.r.t. new basis vectors set.

$$(\boldsymbol{A} - \lambda \boldsymbol{I})\,\boldsymbol{u}_1 = \boldsymbol{0} \implies \boldsymbol{A}\boldsymbol{u}_1 = \lambda \boldsymbol{u}_1 = \boldsymbol{Q} \begin{bmatrix} \lambda & 0 & 0 & 0 & \cdots & 0 \end{bmatrix}^\top$$

$$\boldsymbol{A}\boldsymbol{u}_2 = \boldsymbol{u}_1 + \lambda \boldsymbol{u}_2 = \boldsymbol{Q} \begin{bmatrix} 1 & \lambda & 0 & 0 & \cdots & 0 \end{bmatrix}^\top$$

$$\boldsymbol{A}\boldsymbol{u}_3 = \boldsymbol{u}_2 + \lambda \boldsymbol{u}_3 = \boldsymbol{Q} \begin{bmatrix} 0 & 1 & \lambda & 0 & \cdots & 0 \end{bmatrix}^\top$$

$$\boldsymbol{A}\boldsymbol{u}_4 = \boldsymbol{u}_3 + \lambda \boldsymbol{u}_4 = \boldsymbol{Q} \begin{bmatrix} 0 & 0 & 1 & \lambda & \cdots & 0 \end{bmatrix}^\top$$

$$\hat{\boldsymbol{A}} = \begin{bmatrix} \lambda & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \lambda & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \lambda & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \lambda & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \lambda & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \lambda & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \lambda & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \lambda \end{bmatrix}$$

- **Geometric multiplicity**: Dimension of null space of $(\boldsymbol{A} - \lambda \boldsymbol{I})$
- **Algebraic multiplicity**: Sum of orders of Jordan blocks associated with λ



Example

- Consider a matrix

$$\boldsymbol{A} = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 2 \end{bmatrix}$$

- Eigenvalues: 1, 1, 2
- Eigenvector corresponding to $\lambda_3 = 2$: $[5\ 3\ 1]^\top$
- $\rho(\boldsymbol{A} - \lambda_1 \boldsymbol{I}) = 2$, only one eigenvector w.r.t. λ_1

$$\boldsymbol{B} \triangleq (\boldsymbol{A} - \lambda_1 \boldsymbol{I}) = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & 1 \end{bmatrix}; \quad \boldsymbol{B}^2 = \begin{bmatrix} 0 & 0 & 5 \\ 0 & 0 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

- As $\rho(\boldsymbol{B}^2) = 1 = n - m$, we stop here.
- Look for a $\boldsymbol{v}$ s.t. $\boldsymbol{B}^2\boldsymbol{v} = \boldsymbol{0}$ but $\boldsymbol{B}\boldsymbol{v} \neq \boldsymbol{0}$
- $\boldsymbol{v} = [0\ 1\ 0]^\top$, thus $\boldsymbol{B}\boldsymbol{v} = [1\ 0\ 0]^\top$
- Generalized vector of grade 2

$$\boldsymbol{v}_2 = \boldsymbol{v} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \boldsymbol{v}_1 \triangleq \boldsymbol{B}\boldsymbol{v} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$



Example

- i^{th} column of $\hat{\boldsymbol{A}}$: Representation of $\boldsymbol{A}\boldsymbol{v}_i$ w.r.t. basis $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$:

$$\boldsymbol{A}\boldsymbol{v}_1 = \lambda_1 \boldsymbol{v}_1, \;\; \boldsymbol{A}\boldsymbol{v}_2 = \boldsymbol{v}_1 + \lambda_1 \boldsymbol{v}_2, \;\; \boldsymbol{A}\boldsymbol{v}_3 = \lambda_3 \boldsymbol{v}_3$$

$$\hat{\boldsymbol{A}} = \begin{bmatrix} \lambda_1 & 1 & 0 \\ 0 & \lambda_1 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

- Alternate approach:

$$\hat{\boldsymbol{A}} = \boldsymbol{Q}^{-1}\boldsymbol{A}\boldsymbol{Q} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}; \quad \boldsymbol{Q} = \begin{bmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \boldsymbol{v}_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

- How to define the function of a matrix?

$$\boldsymbol{A}^k \triangleq \boldsymbol{A}\boldsymbol{A}\cdots\boldsymbol{A} \; (k \text{ terms}), \quad \boldsymbol{A}^0 \triangleq \boldsymbol{I}$$

$$f(\lambda) = \lambda^2 + 2\lambda + 1 \implies \boxed{f(\boldsymbol{A}) = \boldsymbol{A}^2 + 2\boldsymbol{A} + \boldsymbol{I}}$$



Minimal Polynomial of Matrix

- Every square matrix mapping $(\mathbb{C}^n, \mathbb{C})$ to itself has a Jordan form.

$$\boldsymbol{A}^k = \boldsymbol{Q}\hat{\boldsymbol{A}}^k\boldsymbol{Q}^{-1}, \quad f(\boldsymbol{A}) = \boldsymbol{Q} f\left(\hat{\boldsymbol{A}}\right)\boldsymbol{Q}^{-1}, \quad f\left(\hat{\boldsymbol{A}}\right) = \boldsymbol{Q}^{-1} f(\boldsymbol{A})\,\boldsymbol{Q}$$

$$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{A}_1 & 0 \\ 0 & \boldsymbol{A}_2 \end{bmatrix}, \quad f(\boldsymbol{A}) = \begin{bmatrix} f(\boldsymbol{A}_1) & 0 \\ 0 & f(\boldsymbol{A}_2) \end{bmatrix}, \quad \boldsymbol{A}^k = \begin{bmatrix} \boldsymbol{A}_1^k & 0 \\ 0 & \boldsymbol{A}_2^k \end{bmatrix}$$

The minimal polynomial of a matrix $\boldsymbol{A}$ is the monic polynomial $\psi(\lambda)$ of least degree such that $\psi(\boldsymbol{A}) = \boldsymbol{0}$.

- $f(\boldsymbol{A}) = \boldsymbol{0} \iff f\left(\hat{\boldsymbol{A}}\right) = \boldsymbol{0}$.
- Similar matrices have the same minimal polynomial.
- Consider distinct eigenvalues of $\boldsymbol{A}$, $\{\lambda_i\}$, with their multiplicities, $n_i \; \forall 1, \ldots, m$.
- Characteristic polynomial: $\Delta(\lambda) \triangleq \det(\lambda\boldsymbol{I} - \boldsymbol{A}) = \Pi_{i=1}^{m}(\lambda - \lambda_i)^{n_i}$

$$\hat{\boldsymbol{A}} = \begin{bmatrix} \hat{\boldsymbol{A}}_1 & \boldsymbol{0} & \cdots & \boldsymbol{0} \\ \boldsymbol{0} & \hat{\boldsymbol{A}}_2 & \cdots & \boldsymbol{0} \\ \vdots & \vdots & \vdots & \vdots \\ \boldsymbol{0} & \cdots & \boldsymbol{0} & \hat{\boldsymbol{A}}_m \end{bmatrix}$$



Minimal Polynomial of Matrix

The largest order of Jordan blocks associated with λ_i is called the index $(\bar{n}_i)$ of eigenvalue λ_i in $\boldsymbol{A}$.

The minimal polynomial of $\boldsymbol{A}$ is given by $\psi(\lambda) = \Pi_{i=1}^{m} (\lambda - \lambda_i)^{\bar{n}_i}$

$$\begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 3 & 1 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 3 & 1 & 0 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Characteristic polynomial: $\Delta(\lambda) = (\lambda - 3)^3(\lambda - 1)$
- Minimal polynomial

$$\psi(\lambda) = (\lambda - 3)(\lambda - 1), \quad (\lambda - 3)^2(\lambda - 1), \quad (\lambda - 3)^3(\lambda - 1)$$

- Characteristic polynomial is always divisible by the minimal polynomial.



Cayley-Hamilton Theorem

Let $\Delta(\lambda) = \det(\lambda \boldsymbol{I} - \boldsymbol{A}) \triangleq \lambda^n + \alpha_1 \lambda^{n-1} + \cdots + \alpha_{n-1}\lambda + \alpha_n$ be the characteristic polynomial of $\boldsymbol{A}$. Then

$$\Delta(\boldsymbol{A}) = \boldsymbol{A}^n + \alpha_1 \boldsymbol{A}^{n-1} + \cdots + \alpha_{n-1}\boldsymbol{A} + \alpha_n \boldsymbol{I} = \boldsymbol{0}$$

Let $\lambda_1, \lambda_2, \ldots, \lambda_m$ be the distinct eigenvalues of $\boldsymbol{A}$ with indices $\bar{n}_1, \bar{n}_2, \ldots, \bar{n}_m$. Let f and g be the two polynomials. Then, the following are equivalent.

- $f(\boldsymbol{A}) = g(\boldsymbol{A})$.
- Either $f = h_1\psi + g$ or $g = h_2\psi + f$, with ψ as minimal polynomial of $\boldsymbol{A}$ and h_1, h_2 are some polynomials.
- $f^l(\lambda_i) = g^l(\lambda_i) \, \forall \, l = 0, 1, 2, \ldots, \bar{n}_i - 1; i = 1, 2, \ldots, m$

  $$f^l(\lambda_i) \triangleq \left.\frac{d^l f(\lambda)}{d\lambda^l}\right|_{\lambda=\lambda_1}, \; g^l(\lambda_i) \triangleq \left.\frac{d^l g(\lambda)}{d\lambda^l}\right|_{\lambda=\lambda_1}$$

Minimal polynomial are usually difficult to compute.



Function of Square Matrix

Let the characteristic polynomial of $\boldsymbol{A}$ be

$$\Delta(\lambda) \triangleq \det(\lambda \boldsymbol{I} - \boldsymbol{A}) = \Pi_{i=1}^{m}(\lambda - \lambda_i)^{n_i}$$

Let f and g be the two arbitrary polynomials. If $f^l(\lambda_i) = g^l(\lambda_i) \forall\ l = 0, 1, 2, \ldots, n_i - 1, i = 1, 2, \cdots, m$ then $f(\boldsymbol{A}) = g(\boldsymbol{A})$.

- Set of n numbers $f^l(\lambda_i)$: Values of f on the spectrum of $\boldsymbol{A}$
- Two polynomials that have the same value on the spectrum of $\boldsymbol{A}$ define the same matrix function.
- Given n numbers, if we can construct a polynomial that gives these numbers on the spectrum of $\boldsymbol{A}$, the polynomial uniquely defines a matrix-valued function of $\boldsymbol{A}$.
- Possibility to construct polynomial of degree $n-1$ that gives these numbers at some preassigned λ.
- For a given $n \times n$ matrix, any polynomial $f(\lambda)$ can be expressed as

$$f(\boldsymbol{A}) = g(\boldsymbol{A}) = \alpha_0 \boldsymbol{I} + \alpha_1 \boldsymbol{A} + \cdots + \alpha_{n-1} \boldsymbol{A}^{n-1}$$



Function of Square Matrix: Example

- If $\boldsymbol{A}$ is of order n, the polynomial $g(\lambda)$ can be any suitably chosen polynomial of degree $n-1$, with n independent parameters.

Compute $f(\boldsymbol{A})$ with $f(\lambda) = \lambda^{100}$ for

$$\boldsymbol{A} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

- Characteristic equation: $\Delta(\lambda) = (\lambda - 1)^2$
- Polynomial of degree $(n-1)$: $g(\lambda) = \alpha_0 + \alpha_1 \lambda$
- Constraints

$$f(1) = g(1), \quad f'(1) = g'(1)$$

$$\alpha_0 + \alpha_1 = 1^{100}, \quad \alpha_1 = 100, \alpha_0 = -99$$

$$\boldsymbol{A}^{100} = \alpha_0 \boldsymbol{I} + \alpha_1 \boldsymbol{A} = -99 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + 100 \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 200 \\ 0 & 1 \end{bmatrix}$$



Function of Square Matrix

Let $f(\lambda)$ be a function (not necessarily a polynomial) that is defined on the spectrum of $\boldsymbol{A}$. If $g(\lambda)$ is a polynomial that has the same values as $f(\lambda)$ on the spectrum of $\boldsymbol{A}$, then the matrix-valued function $f(\boldsymbol{A})$ is defined as $f(\boldsymbol{A}) \triangleq g(\boldsymbol{A})$.

- Any function of $n \times n$ matrix $\boldsymbol{A}$ can be expressed as

$$f(\boldsymbol{A}) = \alpha_0 \boldsymbol{I} + \alpha_1 \boldsymbol{A} + \alpha_2 \boldsymbol{A}^2 + \alpha_3 \boldsymbol{A}^3 + \cdots + \alpha_1 \boldsymbol{A}^{n-1}$$

- Compute characteristic polynomial

$$\Delta(\lambda) = \Pi_{i=1}^{m} (\lambda - \lambda_i)^{n_i}$$

- Assume a polynomial such as

$$g(\lambda) = \alpha_0 + \alpha_1 \lambda_1 + \alpha_2 \lambda^2 + \cdots + \alpha_{n-1} \lambda^{n-1}$$

- Use n equations to compute these coefficients, over the spectrum of $\boldsymbol{A}$.



Function of Square Matrix: Example

Compute $e^{\boldsymbol{A}_1 t}$ for

$$\boldsymbol{A}_1 = \begin{bmatrix} 0 & 0 & -2 \\ 0 & 1 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

- Characteristic equation: $\Delta(\lambda) = (\lambda - 1)^2(\lambda - 2)$
- Polynomial of degree $n - 1$: $g(\lambda) = \alpha_0 + \alpha_1\lambda + \alpha_2\lambda^2$
- Constraints

$$\boxed{f(1) = g(1), \quad f'(1) = g'(1), \quad f(2) = g(2)}$$

$$\alpha_0 + \alpha_1 + \alpha_2 = e^t, \quad \alpha_1 + 2\alpha_2 = te^t, \quad \alpha_0 + 2\alpha_1 + 4\alpha_2 = e^{2t}$$

$$\boxed{\alpha_0 = -2te^t + e^{2t}, \quad \alpha_1 = 2e^t + 3te^t - 2e^{2t}, \quad \alpha_2 = e^{2t} - e^t - te^t}$$

$$e^{\boldsymbol{A}_1 t} = \left(-2te^t + e^{2t}\right)\boldsymbol{I} + \left(2e^t + 3te^t - 2e^{2t}\right)\boldsymbol{A}_1 + \left(e^{2t} - e^t - te^t\right)\boldsymbol{A}_1^2$$

$$e^{\boldsymbol{A}_1 t} = \begin{bmatrix} 2e^t - e^{2t} & 0 & 2e^t - 2e^{2t} \\ 0 & e^t & 0 \\ -e^t + e^{2t} & 0 & 2e^{2t} - e^t \end{bmatrix}$$



Function of Square Matrix: Example

Compute $e^{\boldsymbol{A}_2 t}$ for

$$\boldsymbol{A}_2 = \begin{bmatrix} 0 & 0 & -2 \\ 0 & 1 & 0 \\ 1 & -1 & 3 \end{bmatrix}$$

- Characteristic equation: $\Delta(\lambda) = (\lambda - 1)^2(\lambda - 2)$
- Polynomial of degree $n - 1$: $g(\lambda) = \alpha_0 + \alpha_1\lambda + \alpha_2\lambda^2$
- Constraints: $f(1) = g(1), \quad f'(1) = g'(1), \quad f(2) = g(2)$

$$\alpha_0 + \alpha_1 + \alpha_2 = e^t, \quad \alpha_1 + 2\alpha_2 = te^t, \quad e^{2t} = \alpha_0 + 2\alpha_1 + 4\alpha_2$$

$$\boxed{\alpha_0 = -2te^t + e^{2t}, \quad \alpha_1 = 2e^t + 3te^t - 2e^{2t}, \quad \alpha_2 = e^{2t} - e^t - te^t}$$

$$e^{\boldsymbol{A}_2 t} = \left(-2te^t + e^{2t}\right)\boldsymbol{I} + \left(2e^t + 3te^t - 2e^{2t}\right)\boldsymbol{A}_2 + \left(e^{2t} - e^t - te^t\right)\boldsymbol{A}_2^2$$

$$e^{\boldsymbol{A}_2 t} = \begin{bmatrix} 2e^t - e^{2t} & 2te^t & 2e^t - 2e^{2t} \\ 0 & e^t & 0 \\ -e^t + e^{2t} & -te^t & 2e^{2t} - e^t \end{bmatrix}$$



Function of Square Matrix: Example

- How to compute $f(\hat{\boldsymbol{A}})$ for the matrix in Jordan block form below?

$$\hat{\boldsymbol{A}} = \begin{bmatrix} \lambda_1 & 1 & 0 & \cdots & 0 \\ 0 & \lambda_1 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ 0 & 0 & 0 & \cdots & \lambda_1 \end{bmatrix}$$

- Characteristic equation: $\Delta(\lambda) = (\lambda - \lambda_1)^n$
- Polynomial of degree $n-1$:

$$g(\lambda) = \alpha_0 + \alpha_1(\lambda - \lambda_1) + \alpha_2(\lambda - \lambda_1)^2 + \cdots + \alpha_n(\lambda - \lambda_1)^n$$

- Constraints: $f(1) = g(1), \quad f'(1) = g'(1), \cdots$

$$\alpha_0 = f(\lambda_1), \ \alpha_1 = f'(\lambda_1), \cdots, \alpha_n = \frac{f^{n-1}(\lambda_1)}{(n-1)!}$$

$$f(\hat{\boldsymbol{A}}) = g(\hat{\boldsymbol{A}}) = f(\lambda_1)\boldsymbol{I} + \frac{f^{'}(\lambda_1)}{1!}(\hat{\boldsymbol{A}} - \lambda_1\boldsymbol{I}) + \cdots + \frac{f^{n-1}(\lambda_1)}{(n-1)!}(\hat{\boldsymbol{A}} - \lambda_1\boldsymbol{I})^{n-1}$$



Function of Square Matrix

Let the power series representation of a function f be $f(\lambda) = \sum_{i=0}^{\infty} \alpha_i \lambda^i$ with radius of convergence ρ. Then, the function f of a square matrix $\boldsymbol{A}$ is defined as

$$f(\boldsymbol{A}) \triangleq \sum_{i=0}^{\infty} \alpha_i \boldsymbol{A}^i$$

if the absolute value of all the eigenvalues of $\boldsymbol{A}$ are smaller than ρ, or $\boldsymbol{A}$ has property $\boldsymbol{A}^k = \boldsymbol{0}$ for some integer k.

- Definition meaningful if the infinite series converges.
- If $\boldsymbol{A}^k = \boldsymbol{0}$ for some positive k, then $f(\lambda) = \sum_{i=0}^{k-1} \alpha_i \lambda^i$

$$e^{\lambda t} = 1 + \lambda t + \frac{\lambda^2 t^2}{2!} + \cdots + \frac{\lambda^n t^n}{n!} + \cdots \implies e^{\boldsymbol{A}t} = \sum_{k=0}^{\infty} \frac{\boldsymbol{A}^k t^k}{k!}$$

- Use of infinite series to avoid computation of eigenvalue for using a closed-form matrix expression.



Norms of Vector and Matrix

Any real-valued function of $\boldsymbol{x} \in (\mathbb{C}^n, \mathbb{C})$, denoted by $\|x\|$ can be defined as a norm if it has properties that for any $\boldsymbol{x} \in (\mathbb{C}^n, \mathbb{C})$ and $\alpha \in \mathbb{C}$

- $\|\boldsymbol{x}\| \geq 0$ and $\|\boldsymbol{x}\| = 0$ iff $\boldsymbol{x} = \boldsymbol{0}$.
- $\|\alpha \boldsymbol{x}\| = |\alpha| \|\boldsymbol{x}\|$.
- $\|\boldsymbol{x}_1 + \boldsymbol{x}_2\| \leq \|\boldsymbol{x}_1\| + \|\boldsymbol{x}_2\|$ (Triangular inequality).

- Examples:
  $\|\boldsymbol{x}\|_1 \triangleq \sum_{i=1}^{n} |x_i|, \quad \|\boldsymbol{x}\|_2 \triangleq \left(\sum_{i=1}^{n} |x_i|^2\right)^{1/2}, \quad \|\boldsymbol{x}\|_\infty \triangleq \max_i |x_i|$
- $\|\boldsymbol{x}\|$ is finite iff all the components of $\boldsymbol{x}$ are finite.
- Norm of a matrix, induced norm, with $\boldsymbol{x} \in (\mathbb{C}^n, \mathbb{C})$

$$\|\boldsymbol{A}\| \triangleq \sup_{\boldsymbol{x} \neq \boldsymbol{0}} \frac{\|\boldsymbol{A}\boldsymbol{x}\|}{\|\boldsymbol{x}\|} = \sup_{\|\boldsymbol{x}\|=1} \|\boldsymbol{A}\boldsymbol{x}\|, \quad \|\boldsymbol{A}\boldsymbol{x}\| \leq \|\boldsymbol{A}\| \|\boldsymbol{x}\|$$

$$\|\boldsymbol{A}\|_1 = \max_j \left(\sum_{i=1}^{n} |a_{ij}|\right), \; \|\boldsymbol{A}\|_2 = \sqrt{(\lambda_{\max}(\boldsymbol{A}^\star \boldsymbol{A}))}, \; \|\boldsymbol{A}\|_\infty = \max_i \left(\sum_{j=1}^{n} |a_{ij}|\right)$$

$$\boxed{\|\boldsymbol{A} + \boldsymbol{B}\| \leq \|\boldsymbol{A}\| + \|\boldsymbol{B}\|, \|\boldsymbol{A}\boldsymbol{B}\| \leq \|\boldsymbol{A}\| \|\boldsymbol{B}\|}$$



Inner product of two vectors $\boldsymbol{x}, \boldsymbol{y} \in (\mathbb{C}^n, \mathbb{C})$ is a complex number, denoted by $\langle \boldsymbol{x}, \boldsymbol{y} \rangle$, having the properties that for any $\alpha_1, \alpha_2 \in \mathbb{C}$,

- $\overline{\langle \boldsymbol{x}, \boldsymbol{y} \rangle} = \langle \boldsymbol{y}, \boldsymbol{x} \rangle$
- $\langle \alpha_1 \boldsymbol{x}_1 + \alpha_2 \boldsymbol{x}_2, \boldsymbol{y} \rangle = \bar{\alpha}_1 \langle \boldsymbol{x}_1, \boldsymbol{y} \rangle + \bar{\alpha}_2 \langle \boldsymbol{x}_2, \boldsymbol{y} \rangle$
- $\langle \boldsymbol{x}, \boldsymbol{x} \rangle > 0 \, \forall \boldsymbol{x} \neq \boldsymbol{0}$

- In complex vector space $(\mathbb{C}^n, \mathbb{C})$, $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \boldsymbol{x}^\star \boldsymbol{y} = \sum_{i=1}^{n} \bar{x}_i y_i$

$$\langle \boldsymbol{x}, \boldsymbol{A}\boldsymbol{y} \rangle = \boldsymbol{x}^\star \boldsymbol{A}\boldsymbol{y}, \; \langle \boldsymbol{A}^\star \boldsymbol{x}, \boldsymbol{y} \rangle = (\boldsymbol{A}^\star \boldsymbol{x})^\star \boldsymbol{y} = \boldsymbol{x}^\star \boldsymbol{A}\boldsymbol{y} \implies \boxed{\langle \boldsymbol{x}, \boldsymbol{A}\boldsymbol{y} \rangle = \langle \boldsymbol{A}^\star \boldsymbol{x}, \boldsymbol{y} \rangle}$$

Schwarz Inequality: If we define $\|\boldsymbol{x}\| = \sqrt{\langle \boldsymbol{x}, \boldsymbol{x} \rangle}$, $|\langle \boldsymbol{x}, \boldsymbol{y} \rangle| \leq \|\boldsymbol{x}\| \|\boldsymbol{y}\|$

- Inequality true for $\boldsymbol{y} = \boldsymbol{0}$. For $\boldsymbol{y} \neq \boldsymbol{0}$

$$0 \leq \langle \boldsymbol{x} + \alpha \boldsymbol{y}, \boldsymbol{x} + \alpha \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \boldsymbol{x} \rangle + \bar{\alpha} \langle \boldsymbol{y}, \boldsymbol{x} \rangle + \alpha \langle \boldsymbol{x}, \boldsymbol{y} \rangle + \alpha \bar{\alpha} \langle \boldsymbol{y}, \boldsymbol{y} \rangle$$

- With $\alpha = -\dfrac{\langle \boldsymbol{y}, \boldsymbol{x} \rangle}{\langle \boldsymbol{y}, \boldsymbol{y} \rangle}$, $\quad \langle \boldsymbol{x}, \boldsymbol{x} \rangle \geq \dfrac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle \langle \boldsymbol{y}, \boldsymbol{x} \rangle}{\langle \boldsymbol{y}, \boldsymbol{y} \rangle} = \dfrac{|\langle \boldsymbol{x}, \boldsymbol{x} \rangle|^2}{\langle \boldsymbol{y}, \boldsymbol{y} \rangle}$.



Positive Definiteness of Matrix

An $n \times n$ real matrix $\boldsymbol{M}$ is said to be symmetric if its transpose equals itself.

A symmetric matrix $\boldsymbol{M}$ is said to be positive definite, denoted by $\boldsymbol{M} > 0$, if $\boldsymbol{x}^\top \boldsymbol{M} \boldsymbol{x} > 0$ for every nonzero $\boldsymbol{x}$. It is positive semidefinite, denoted by $\boldsymbol{M} \geq 0$, if $\boldsymbol{x}^\top \boldsymbol{M} \boldsymbol{x} \geq 0$ for every nonzero $\boldsymbol{x}$.

- Every eigenvalue of $\boldsymbol{M}$ is positive (zero or positive).
- All the leading principal minors of $\boldsymbol{M}$ are positive (all the principal minors of $\boldsymbol{M}$ are zero or positive).

An $m \times n$ matrix $\boldsymbol{H}$, with $m \geq n$, has rank n, if and only if the $n \times n$ matrix $\boldsymbol{H}^\top \boldsymbol{H}$ has rank n or $\det(\boldsymbol{H}^\top \boldsymbol{H}) \neq 0$. An $m \times n$ matrix $\boldsymbol{H}$, with $m \leq n$, has rank m, if and only if the $m \times m$ matrix $\boldsymbol{H}\boldsymbol{H}^\top$ has rank m or $\det(\boldsymbol{H}\boldsymbol{H}^\top) \neq 0$.



Reference

Reference

- **Chen, Chi-Tsong, "Linear System Theory and Design", CBS College Publishing, 1984.**

